VeggieCare: An AI- and IoT-Based Smart Vegetable Farm Management, E-Commerce, and Harvest Distribution System

Mindoro State University, Calapan City Campus

Oct 2026

TASK	Start wk	End wk
1 REQUIREMENTS GATHERING AND ANALYSIS		
1.1 Project planning	1	6
1.2 Needs assessment	1	6
1.3 Project scope definition	1	6
1.4 Client consultation	7	9
1.5 Requirements specification	7	15
1.6 Implementation plan	7	15
2 DESIGN PHASE		
2.1 System analysis and design	10	15
2.2 System architecture and diagrams	10	15
2.3 Database schema design	10	15
2.4 Sensor and irrigation wiring design	10	15
2.5 User interface design	10	15
3 DEVELOPMENT		
3.1 Hardware setup and sensor calibration	16	40
3.2 AI dataset collection and model training	16	40
3.3 Iteration 1: login, roles, sensor readings	16	40
3.4 Iteration 2: irrigation and weather forecast	16	40
3.5 Iteration 3: disease scan and fertilizer	16	40
3.6 Iteration 4: e-commerce, orders, Maya sandbox	16	40
3.7 Iteration 5: harvest, demand matching, delivery	16	40
4 TESTING AND VALIDATION		
4.1 Unit testing	41	48
4.2 AI model accuracy testing	41	48
4.3 Integration testing	41	48
4.4 Functionality testing	41	48
4.5 User testing	41	48
5 DEPLOYMENT AND IMPLEMENTATION		
5.1 System hosting	49	52
5.2 Installation in pilot zones	49	52
5.3 User training (owner, staff, buyers)	49	52
6 EVALUATION AND CONTINUOUS IMPROVEMENT		
6.1 System evaluation (ISO/IEC 25010)	53	57
6.2 Feedback collection	53	57
6.3 System refinement	53	57
6.4 Final documentation and defense preparation	56	60

